

OpenStax Calculus Volume 1

Section 4.6 Exercises: Limits at Infinity and Asymptotes

Source: <https://openstax.org/books/calculus-volume-1/pages/4-6-limits-at-infinity-and-asymptotes>.
Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

For the following exercises, examine the graphs. Identify where the vertical asymptotes are located.

1. *[Figure: The function graphed decreases very rapidly as it approaches $x = 1$ from the left, and on the other side of $x = 1$, it seems to start near infinity and then decrease rapidly.]*
2. *[Figure: The function graphed increases very rapidly as it approaches $x = -3$ from the left, and on the other side of $x = -3$, it seems to start near negative infinity and then increase rapidly to form a sort of U shape that is pointing down, with the other side of the U being at $x = 2$. On the other side of $x = 2$, the graph seems to start near infinity and then decrease rapidly.]*
3. *[Figure: The function graphed decreases very rapidly as it approaches $x = -1$ from the left, and on the other side of $x = -1$, it seems to start near negative infinity and then increase rapidly to form a sort of U shape that is pointing down, with the other side of the U being at $x = 2$. On the other side of $x = 2$, the graph seems to start near infinity and then decrease rapidly.]*
4. *[Figure: The function graphed decreases very rapidly as it approaches $x = 0$ from the left, and on the other side of $x = 0$, it seems to start near infinity and then decrease rapidly to form a sort of U shape that is pointing up, with the other side of the U being at $x = 1$. On the other side of $x = 1$, there is another U shape pointing down, with its other side being at $x = 2$. On the other side of $x = 2$, the graph seems to start near negative infinity and then increase rapidly.]*
5. *[Figure: The function graphed decreases very rapidly as it approaches $x = 0$ from the left, and on the other side of $x = 0$, it seems to start near infinity and then decrease rapidly to form a sort of U shape that is pointing up, with the other side being a normal function that appears as if it will take the entirety of the values of the x -axis.]*

For the following functions $f(x)$, determine whether there is an asymptote at $x = a$. Justify your answer without graphing on a calculator.

6. $f(x) = \frac{x+1}{x^2+5x+4}, a = -1$
7. $f(x) = \frac{x}{x-2}, a = 2$
8. $f(x) = (x+2)^{3/2}, a = -2$
9. $f(x) = (x-1)^{-1/3}, a = 1$

10. $f(x) = 1 + x^{-2/5}, a = 1$

For the following exercises, evaluate the limit.

11. $\lim_{x \rightarrow \infty} \frac{1}{3x+6}$

12. $\lim_{x \rightarrow \infty} \frac{2x-5}{4x}$

13. $\lim_{x \rightarrow \infty} \frac{x^2-2x+5}{x+2}$

14. $\lim_{x \rightarrow -\infty} \frac{3x^3-2x}{x^2+2x+8}$

15. $\lim_{x \rightarrow -\infty} \frac{x^4-4x^3+1}{2-2x^2-7x^4}$

16. $\lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2+1}}$

17. $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2-1}}{x+2}$

18. $\lim_{x \rightarrow \infty} \frac{4x}{\sqrt{x^2-1}}$

19. $\lim_{x \rightarrow -\infty} \frac{4x}{\sqrt{x^2-1}}$

20. $\lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x-\sqrt{x+1}}$

For the following exercises, find the horizontal and vertical asymptotes.

21. $f(x) = x - \frac{9}{x}$

22. $f(x) = \frac{1}{1-x^2}$

23. $f(x) = \frac{x^3}{4-x^2}$

24. $f(x) = \frac{x^2+3}{x^2+1}$

25. $f(x) = \sin(x) \sin(2x)$

26. $f(x) = \cos x + \cos(3x) + \cos(5x)$

27. $f(x) = \frac{x \sin(x)}{x^2-1}$

28. $f(x) = \frac{x}{\sin(x)}$

29. $f(x) = \frac{1}{x^3+x^2}$

30. $f(x) = \frac{1}{x-1} - 2x$

31. $f(x) = \frac{x^3+1}{x^3-1}$

32. $f(x) = \frac{\sin x + \cos x}{\sin x - \cos x}$

33. $f(x) = x - \sin x$

34. $f(x) = \frac{1}{x} - \sqrt{x}$

For the following exercises, construct a function $f(x)$ that has the given asymptotes.

35. $x = 1$ and $y = 2$

36. $x = 1$ and $y = 0$

37. $y = 4, x = -1$

38. $x = 0$

For the following exercises, graph the function on a graphing calculator on the window $x = [-5, 5]$ and estimate the horizontal asymptote or limit. Then, calculate the actual horizontal asymptote or limit.

39. [T] $f(x) = \frac{1}{x+10}$

40. [T] $f(x) = \frac{x+1}{x^2+7x+6}$

41. [T] $\lim_{x \rightarrow -\infty} x^2 + 10x + 25$

42. [T] $\lim_{x \rightarrow -\infty} \frac{x+2}{x^2+7x+6}$

43. [T] $\lim_{x \rightarrow \infty} \frac{3x+2}{x+5}$

For the following exercises, draw a graph of the functions without using a calculator. Be sure to notice all important features of the graph: local maxima and minima, inflection points, and asymptotic behavior.

44. $y = 3x^2 + 2x + 4$

45. $y = x^3 - 3x^2 + 4$

46. $y = \frac{2x+1}{x^2+6x+5}$

47. $y = \frac{x^3+4x^2+3x}{3x+9}$

48. $y = \frac{x^2+x-2}{x^2-3x-4}$

49. $y = \sqrt{x^2 - 5x + 4}$

50. $y = 2x\sqrt{16 - x^2}$

51. $y = \frac{\cos x}{x}$, on $x = [-2\pi, 2\pi]$

52. $y = \frac{\sqrt{x^2+2}}{x+1}$

53. $y = x \tan x, x = [-\pi, \pi]$

54. $y = x \ln(x), x > 0$

55. $y = x^2 \sin(x), x = [-2\pi, 2\pi]$

56. For $f(x) = \frac{P(x)}{Q(x)}$ to have an asymptote at $y = 2$ then the polynomials $P(x)$ and $Q(x)$ must have what relation?

57. For $f(x) = \frac{P(x)}{Q(x)}$ to have an asymptote at $x = 0$, then the polynomials $P(x)$ and $Q(x)$ must have what relation?

58. If $f'(x)$ has asymptotes at $y = 3$ and $x = 1$, then $f(x)$ has what asymptotes?

59. Both $f(x) = \frac{1}{(x-1)}$ and $g(x) = \frac{1}{(x-1)^2}$ have asymptotes at $x = 1$ and $y = 0$. What is the most obvious difference between these two functions?

60. True or false: Every ratio of polynomials has vertical asymptotes.